TIAN Zhuang

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AGENT SKILLS

- **Computer Agentic Skills:**
  - ✧ Full-stack engineering developer: Front-End (Vue, React), Back-End (Django, FastApi, Solidity…)
  - ✧ Database: MySQL, MongeDB, PostgreSQL, Redis, Spark…
  - ✧ Could server: Google Cloud, Hostinger…
  - ✧ Familiar with Python, Transformer and Mainstream artificial intelligence algorithms.
  - ✧ Data Visualization: Scikit-learn module built on NumPy, SciPy, Pandas and Matplotlib, Dashboard (Tableau)
  - ✧ Development tools: VSCode, OpenClaw, Qt framework, Anaconda, DevEco, Github and Agent Skills…
- **Languages**: IELTS (6.5), Mandarin (Putonghua Proficiency Test: Level 2-A)

EDUCATION

Lingnan University **2025 - 2026**

Master of Science in Artificial Intelligence and Business Analysis

- **Curriculum**
  - ✧ Practical Application of Deep Learning
  - ✧ Business Data Management
  - ✧ Data Visualization
  - ✧ Principle of Data Analytics and Programming
  - ✧ Programming with Generative AI
  - ✧ Big Data Analytics
  - ✧ Computer Vision
  - ✧ Machine Learning for Business
  - ✧ Blockchain
- **GPA**: 3.6/4.0 (Semester1)

Federation University Australia - Shaoguan University (3+1) **09/2021-06/2025**

Bachelor of Engineering in Computer Science and Technology

Introduction: The undergraduate program is a "3+1" programme run by SGU in collaboration with Federation University Australia. Without going to Australia in the fourth year, I studied at SGU and received major courses provided by teachers from the Federation University Australia.

- **GPA**: 3.49/5.0 (84.94/100)
- **Certificate**: Certificate of National Information Security Test Program (Level 1)
- **Honours & Awards:**
  - ✧ China National Scholarship (11/2024)
  - ✧ First-class Academic Scholarship of the School of Information Engineering (10/2024)
  - ✧ Merit Student of the School of Information Engineering (10/2024)
  - ✧ Outstanding Individual in Research Innovation (10/2024)
  - ✧ Academic Excellent Award (10/2024)

PROJECTS

Multi-Agents LLM Financial Trading Decision Making Framework (Enhance TradingAgents) **02/2026**

Project Description: Built on large language models (LLMs) and multi-agent architecture, this end-to-end automated stock investment research and decision-making system realizes full-process automation from target analysis, multi-perspective debate, trading decision-making to risk control, ultimately generating structured trading signals (BUY/SELL/HOLD).

- The overall system architecture and split multi-agent roles (Market Analyst, Fundamental Researcher, Trader, Risk Controller, Debate Referee) to enhance decision robustness and interpretability through cross-role collaboration.
- Implemented the investment research decision workflow orchestration based on LangGraph, constructing a complete execution graph: Data Acquisition → Multi-dimensional Analysis → Bull-Bear Debate → Trading Proposal → Risk Validation → Signal Output.
- Established a unified LLM adaptation layer, supporting seamless switching between multi-vendor models (OpenAI / Google / Anthropic / Ollama) to improve system compatibility and scalability.
- Encapsulated a financial data toolchain, integrating data sources such as yfinance and Alpha Vantage to enable automatic retrieval and structured parsing of market quotes, news, fundamentals, and market sentiment.
- Designed state management and memory mechanisms: implemented lightweight historical scenario retrieval based on BM25, combined with a reflection mechanism to form an iterable investment research memory loop.
- Completed end-to-end engineering implementation, supporting automatic execution with specified tickers and dates, and outputting comprehensive decision logs and standardized trading signals

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Kronos Multi-Agent Forecasting & Trading (Kronos-based) **01/2026**

Overall: A generative prediction system for financial Time series (OHLCVA) is built around the core of "Tokenizer quantization and discretization + Transformer autoregression generation + multi-pipeline fine-tuning + Web visualization", and it has a complete basic capability of "training - inference - visualization - backtesting".

Core technical architecture:

- ✧ The core of sequence modeling: By combining "binary spherical quantization (BSQuantizer) + hierarchical Transformer", continuous OHLCVA sequences are discretized into tokens and then autoregressive generation is performed, considering both compression efficiency and generation capability. The Predictor layer introduces dedicated time series designs such as RoPE self-attention, cross-attention conditioning, and time embedding to adapt to the time dependence of financial sequences.

Flexible training system:

- ✧ Supports dual pipelines (Qlib professional quantization/CSV universal), multiple training modes (sequential/single component/distributed), and lightweight fine-tuning (LoRA for attention/FFN layer), which can adapt to different data sources, computing power environments and adaptation requirements.

Engineering foundation: Covering multiple devices such as CPU, CUDA, and MPS, it provides Web UI visualization and regression testing to ensure stable output and has an end-to-end basic link from "model training" to "policy backtesting".

The core logic of optimization:

- ✧ Shifting from a "model-centric" approach to a "trade-centric" one—no longer solely pursuing point prediction accuracy, but ensuring prediction results can be transformed into stable, tradable, and verifiable profits. Specifically, outline the phased implementation value and implementation paths.

SRR-Case-Processing-System **11/2025**

Introduction: The Architecture Services Department (ArchSD) in Hong Kong aims to develop an AI-powered support system to enhance the efficiency of slope and tree maintenance workflows.

Overview: An AI-powered document processing system that automatically extracts and classifies case data from Slope Risk Reports (SRR) in multiple file formats. Transform unstructured documents (TXT, TMO PDFs, RCC PDFs) into standardized case records with AI-powered classification and data extraction.

GitHub link: https://github.com/SRR-Project-Team/SRR-Case-Processing-System

Web URL: Private

Key Features:

- ✧ Multi-format Processing: TXT files, TMO PDFs, RCC PDFs
- ✧ AI Classification: 92% accuracy case type, 98% subject matter
- ✧ Historical Case Matching: Search 5,298 historical cases for patterns
- ✧ OCR Processing: Advanced text recognition for scanned documents
- ✧ Batch Processing: Handle multiple files simultaneously
- ✧ Integrated Chat Interface: Popup-based file management
- ✧ Tree Information Tracking: 32,405 trees with location data
- ✧ Location-Slope Learning: 403 auto-learned mappings

This project participated in the 2026 Lingnan Cup, in conjunction with business report analysis, etc

Lingnan University Policy System Using Agentic Retrieval-Augmented Generation (RAG)Framework **12/2025**

Background: The policy documents of Lingnan University are usually lengthy and complex, covering languages such as Cantonese, English and simplified Chinese. Traditional RAG has a lot of illusions, and the existing framework is difficult to fine-tune many details. So, I developed a custom Agentic RAG framework as required by my mentor.

GitHub link：https://github.com/terzotian/RAGagent

Web URL: https://agentrag.app

- Traditional question-answering (QA) systems, which rely on predefined QA pairs or keyword-based retrieval, have limited capacity to interpret complex or novel queries. While Large Language Models (LLMs) can generate coherent, human-like responses, they are prone to hallucinations—producing content not grounded in the source material. In policy and compliance contexts, where factual accuracy and transparency are essential, such errors are unacceptable. Retrieval-Augmented Generation (RAG) offers an effective solution by integrating retrieval and generation components. The retrieval module extracts relevant policy segments from a document repository, while the generation module synthesizes coherent responses grounded in those sources. This hybrid approach enhances factual reliability, mitigates hallucinations, and ensures that outputs remain verifiable and consistent with authoritative policy content.

Project Description:

- This project aims to design, develop, and deploy a Policy Question Answering (QA) System leveraging the Agentic Retrieval-Augmented Generation (RAG) framework, fully implemented within the Google Cloud Platform (GCP)

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environment. The system will be built and evaluated using Lingnan University's policies, enabling users to query university policy documents and receive concise, accurate, and evidence-based responses. The primary objective of this project is developing an AI-driven, cloud-native application that integrates modern machine learning and cloud computing technologies.

- All components—including data storage, retrieval, inference, and deployment—will be designed, developed, tested, and managed entirely within the GCP ecosystem.
- Core Task:
  - ✧ Policy document processing on GCP
  - ✧ Semantic search and query retrieval
  - ✧ RAG model development and integration
  - ✧ RAG model evaluation and performance optimization
  - ✧ Deployment and service delivery on GCP
- Techniques that Involved:
  - ✧ Programming Skills
  - ✧ Knowledge of NLP and Machine Learning
  - ✧ Familiarity with Cloud Services
- Achievement:
  - ✧ Development of an RAG system
  - ✧ Practical experience with NLP pipelines, vector search, and document intelligence
  - ✧ Techniques for evaluating and mitigating hallucinations in generative AI systems
  - ✧ Hands-on training with GCP AI tools
- Remarks:
  - ✧ This project is highly practical in nature, with a strong focus on production launch.
  - ✧ Students will develop skills that are directly relevant to the Hong Kong job market.
  - ✧ External support from experienced AI technicians is also available.

Driver Fatigue Monitoring System Based on Computer Vision **11/2025**

The overall system architecture adopts a three-layer modular structure of "Data - Model - Decision", forming a complete closed loop of "Data Collection - Feature Extraction - Fatigue Prediction - Intelligent Decision-Making - Feedback Optimization", ensuring seamless connection and efficient collaboration among all links.

- The design concept of the architecture: The three-layer architecture performs its own duties while also interacting with each other:
  - ✧ The data layer is responsible for precise collection and feature extraction, providing high-quality input for subsequent analysis.
  - ✧ The model layer is based on the fusion of multiple algorithms to achieve precise prediction of fatigue levels.
  - ✧ The decision layer, making level dynamically optimizes intervention strategies through reinforcement learning to balance safety assurance and driving experience.
- Core workflow data starts from collection. After multi-dimensional features are extracted by MediaPipe, they are input into the random forest or CNN model for fatigue level determination. Then, the Q-Learning algorithm generates personalized intervention instructions based on the determination results. Finally, the intervention is executed through hardware, and the feedback data is fed back to the system. Achieve continuous optimization of the model.

Wildfire Smoke Detection Software Based on YOLOv8 **06/2023**

Part of **"An Intelligent UVA Used for Early Forest Fire Detection"**

Member (responsible for Fire-Smoke generation)

Introduction: The system enables an all-round detection of forest fire by connecting the Himawari-8 data server through the FTP connection pool, a fire lookout tower, and a drone with camera.

- Engaged in team discussions to determine the specific goals of the algorithm, including accuracy requirements and application scenarios
- Referred to similar cases, compared varied algorithms for wildfire smoke detection, selected the YOLOv8 deep learning model, and examined its pros and cons
- Applied the YOLOv8 model to work out the algorithm, including:
  - ✧ Collected datasets containing images of wildfire smoke, preprocessed the collected data and identified relevant features for smoke detection
  - ✧ Defined model parameters, trained the model with the annotated data, and optimized its performance by adjusting hyperparameters
  - ✧ Assessed the accuracy and generation capability of the model and modified it based on validation results
  - ✧ Evaluated the final model to measure its performance in actual scenarios
- *Achievement*: The application for software copyright was submitted to the China Copyright Protection Center.

A Front-End and Back-End Separated System Based on Seafood Price Prediction **04/2025**

- My undergraduate thesis was a front-end and back-end separated project developed based on the Vue and Django

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frameworks. Using Jupyter notebook, I crawled and cleaned the aquatic product data from Wanbang International Data website and saved it in the Navicat database.

- The web system's function is to provide registered users with a list of aquatic products sorted by time. The administrator is responsible for managing users and products. Users can interactively view the price changes of different aquatic products in each province, the price comparisons among products, and the visualization of the origin and the quantity of each province; with the help of machine learning, they can predict price trends.

A Personal Resume Website **06/2024**

- Determined the placement and styling of the website
- Used VS Code for editing HTML, CSS, JavaScript and PHP codes to generate dynamic displays and adaptive layouts
- Uploaded website codes onto the Hostinger server to realize online connection and gain a domain name and an email account, enabling user access and contact
- *Link*: www.terzotian.com

A Decision Tree Model for Prediction **05/2024**

- Accessed the Titanic survival dataset online and preprocessed the dataset with Jupyter notebook, including deleting irrelevant data and converting the non-numeric data to numeric data
- Created new features using feature engineering techniques and presented data with visual charts or tables
- Selected suitable algorithms to train the model and adjusted hyperparameters to improve model performance

COMPETITIONS

University Computer Games Championship

➢ **Gomoku Competition,** *Leader* **05/2024-08/2024**

Supervisor: Dr. Jiang Changjin

Introduction: The University Computer Games Championship includes traditional games, such as chess, Go, and Gomoku. Gomoku competitors are required to develop an AI program for playing Gomoku to match against others.

- Led discussions to determine the required functions of the program and search algorithms needed for program development, and planned the design of the evaluation function and user interface
- Mastered Gomoku rules and conceived how to design opening functions in accordance with Yamaguchi opening rule
- Applied Alpha-Beta pruning and heuristic search to increase the efficiency of identifying optimal moves
- Tested and improved the C++program in Qt Creator, perfected the user interface, and optimized opening functions to provide professional game experiences
- *Achievement*: won the national first prize with the AI Gomoku software designed with a 15×15 Go board

➢ **No Go Competition,** *Member* **05/2024-08/2024**

Introduction: No Go Competition in the Championship, completely different from normal Gomoku games, allows competitors to place stones in turns. The aim is not to eat the other player's stones, or otherwise the player placing stones lose a score. Suicide and pass are prohibited, and there is no draw in this competition.

Supervisor: Dr. Jiang Changjin

- Tested and modified possible program failures
- Prepared the technical report on the gaming software, especially the UB-tree, UCB algorithm, and move generator used in the software program
- *Achievement*: won the national first prize with the AI gaming software for No Go designed with a 9×9 Go board

INTERNSHIPS

Synnex Technology International Corporation (Shenzhen) **09/2024-01/2025**

Huawei Datacom Pre-sales Engineer

- **Wi-Fi deployment advising**: Recommended specific AP types and provided deployment solutions based on customers' requirements; optimized frequency band allocation by using dual-band (2.4GHz and 5GHz) APs, configured SSID segregation to enable separate network access permissions for employees and guests, reserved 20% of AP interfaces to support future business expansion, recommended using PoE switches to simplify AP wiring
- **Technical writing & consultation**: organized technical documents (configuration guides, user manuals), analyzed Huawei datacom products (routers, switches, APs), provided technical consultation on Huawei's datacom products, and assist customers in choosing the right products and solutions

Bank of Communications, Zhuhai **07/2024-08/2024**

Customer Service Manager Assistant

- Organized customers' information and materials and inputted them into the bank's computer

Zhuhai New Horizon Communication Development Co., Ltd. **07/2022-08/2022**

Intern

- Reviewed and processed orders of goods, tracked and updated order status, and recorded transactions
- Maintained good relationships with customers and provided after-sales services

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QUALIFICATION

Huawei Developers **08/2024**

- Successfully completed the online course on Harmony OS application development and received a pass

Cisco Systems, Inc. **11/2022**

- Attended online lectures pertinent to cybersecurity, including cybersecurity administration, threat analysis, system safeguards, network defence, resource specialist

ACTIVITIES

Class Monitor **09/2021-06/2025**

- Assisted with managing student affairs
- Student Member of Chinese Association for Artificial Intelligence
- *Achievement*: Awarded the title of "Outstanding Class Monitor"